Type MS · Find named items in a program

Answer 1 of 18 · Question 1 part d.iv · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part d.iv in this mark scheme extract

1(a)	A1	**The only correct answer is B** ***A*** *is not correct because for is used in iteration* ***C*** *is not correct because new is used to declare an object* ***D*** *is not correct because while is used in repetition*		**1**

Question	mp	Answer	Additional Guidance	Mark
1(b)(i)	B1 B2 B3	Award one mark for each correct cell: **Description** / **Syntax error** / **Logic error** / **Runtime error** The program does not produce the expected output. / / ✔ / The program does not translate. / ✔ / / The program crashes during execution. / / / ✔		**3**

Type MS · Find named items in a program

Answer 1 of 18 · Question 1 part d.iv · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part d.iv in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(b)(ii)	B4	Line 15 - : added to end of line		
	B5	Line 20 – 0 changed to 1		
	B6	Line 22 – missing comma added between string and greenCount		3

```
# Q01bii

# initialise counters
ballCount = 0
greenCount = 0
redCount = 0

ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
             "green", "green", "green", "red", "green", "red", "green", "red",
             "red", "green", "green", "red", "red", "green", "green", "red",
             "green", "red", "red", "green", "green", "red", "green", "green",
             "red", "green", "green", "green", "red", "green", "red","green"]

# iterate over ball array
for ball in ballArray:
    ballCount = ballCount + 1
    if ball == "green":
        greenCount = greenCount +1
    else:
        redCount = redCount + 1

print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
      "balls \n", "Total: ", ballCount, "balls")
```

2 of 23

Type MS · Find named items in a program

Answer 1 of 18 · Question 1 part d.iv · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part d.iv in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(c)(i)	C1	Award one mark for: • To store a value that must not be changed/prevent accidental change of value • To simplify code maintenance/so that if the value must be updated the change only has to made in a single place • To avoid inconsistency error of specifying same value multiple times • To make code more readable/give meaningful names to values	Accidental change cannot refer to 'user'	1

Question	mp	Answer	Additional guidance	Mark
1(c)(ii)	C2 C3	Award **two** marks for a linked description such as: • A variable stores a single value (1) whereas an array stores multiple values (1)		2

Question	mp	Answer	Additional guidance	Mark
1(d)(i)	D1	Award **one** mark for: • getWages (1)	Accept 'main' Accept: def getWages def getWages (inSales, inAge) line 7 (Python)	1

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Type MS · Find named items in a program

Answer 1 of 18 · Question 1 part d.iv · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part d.iv in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(d)(ii)	D2	Award **one** mark for: • and (1)	Do not accept line numbers Accept: &&	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iii)	D3	Award **one** mark for any of the following: • sales (1) • wages (1) • inSales (1)	Do not accept line numbers	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iv)	D4	Award **one** mark for any of the following: • inSales (1) • inAge (1)	Do not accept line numbers	1

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Type MS · Find named items in a program

Answer 1 of 18 · Question 1 part d.iv · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part d.iv in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 1 of 13 · find part c.iii in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 2 of 13 · find part c.iii in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 3 of 13 · find part c.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 4 of 13 · find part c.iii in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 5 of 13 · find part c.iii in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 6 of 13 · find part c.iii in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 7 of 13 · find part c.iii in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 8 of 13 · find part c.iii in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 9 of 13 · find part c.iii in this mark scheme extract

Q2(c)(iv)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 10 of 13 · find part c.iii in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 11 of 13 · find part c.iii in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 12 of 13 · find part c.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 13 of 13 · find part c.iii in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 1 of 13 · find part c.iv in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 2 of 13 · find part c.iv in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 3 of 13 · find part c.iv in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 4 of 13 · find part c.iv in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 5 of 13 · find part c.iv in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 6 of 13 · find part c.iv in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 7 of 13 · find part c.iv in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 8 of 13 · find part c.iv in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 9 of 13 · find part c.iv in this mark scheme extract

Q2(c)(iv)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 10 of 13 · find part c.iv in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 11 of 13 · find part c.iv in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 12 of 13 · find part c.iv in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 13 of 13 · find part c.iv in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 1 of 10 · find part e.iii in this mark scheme extract

1 (a)	A1	**C** (equals) (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 2 of 10 · find part e.iii in this mark scheme extract

1 (b)	B1	Award one mark for any of: • TRUE **and** FALSE (1) • Yes **and** No (1) • T **and** F (1) • 0 **and** 1 (1)	Ignore capitalisation	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 3 of 10 · find part e.iii in this mark scheme extract

1 (d)	D1	Award one mark for any of: • (Symbolic name associated with) a value that may be changed (1) • A container used to store data (1) • A data store whose contents can change whilst a program is executing (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 4 of 10 · find part e.iii in this mark scheme extract

1 (e) (i)	E1	Python 6 C# 12 Java 8		(1)

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 5 of 10 · find part e.iii in this mark scheme extract

1 (e) (ii)	E2	Python 3 C# 7 Java 3		(1)

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 6 of 10 · find part e.iii in this mark scheme extract

1 (e) (iii)	E3	Award one mark for any of: • amount (1) • x (1) • total (1) • count (1)	Accept listNumbers or numArray	(1)

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 7 of 10 · find part e.iii in this mark scheme extract

1 (e) (iv)	E4	message		(1)

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 8 of 10 · find part e.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 9 of 10 · find part e.iii in this mark scheme extract

1 (c)	C1	2.1.5	Delete space in variable name (1)		(3)
	C2	2.1.5	Capitalise O in variable name (1)		
	C3	2.1.5	Correction to print statement (1)		
Code example					
Python					

```
numOne = 25
numTwo = 36
numThree = numOne * numTwo
print (numThree)
```

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Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 10 of 10 · find part e.iii in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 1 of 10 · find part e.iv in this mark scheme extract

1 (a)	A1	**C** (equals) (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 2 of 10 · find part e.iv in this mark scheme extract

1 (b)	B1	Award one mark for any of: • TRUE **and** FALSE (1) • Yes **and** No (1) • T **and** F (1) • 0 **and** 1 (1)	Ignore capitalisation	(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 3 of 10 · find part e.iv in this mark scheme extract

1 (d)	D1	Award one mark for any of: • (Symbolic name associated with) a value that may be changed (1) • A container used to store data (1) • A data store whose contents can change whilst a program is executing (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 4 of 10 · find part e.iv in this mark scheme extract

1 (e) (i)	E1	Python 6 C# 12 Java 8		(1)

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 5 of 10 · find part e.iv in this mark scheme extract

1 (e) (ii)	E2	Python 3 C# 7 Java 3		(1)

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 6 of 10 · find part e.iv in this mark scheme extract

1 (e) (iii)	E3	Award one mark for any of: • amount (1) • x (1) • total (1) • count (1)	Accept listNumbers or numArray	(1)

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 7 of 10 · find part e.iv in this mark scheme extract

1 (e) (iv)	E4	message		(1)

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Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 8 of 10 · find part e.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 9 of 10 · find part e.iv in this mark scheme extract

1 (c)	C1	2.1.5	Delete space in variable name (1)		(3)
	C2	2.1.5	Capitalise O in variable name (1)		
	C3	2.1.5	Correction to print statement (1)		
Code example					
Python					

```
numOne = 25
numTwo = 36
numThree = numOne * numTwo
print (numThree)
```

9 of 16

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 10 of 10 · find part e.iv in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 1 of 8 · find part d.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (b)	B1	Award up to 2 marks for a linked explanation such as: • The number of keys are limited/only one shift used (1) making it easy to use brute force to decrypt (1) • It can be easy to find commonly used letters (e.g. E) (1) and guess the key (1)	Accept alternative similar wording.	(2)

Type MS · Find named items in a program

Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 2 of 8 · find part d.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (c)		Award 1 mark each up to a maximum of 4 for: <table><tr><td>Encrypted letter</td><td>f</td><td>l</td><td>m</td><td>k</td><td>t</td><td>r</td><td>w</td><td>h</td><td>e</td><td>e</td></tr><tr><td>Keyword letter</td><td>t</td><td>h</td><td>i</td><td>r</td><td>t</td><td>y</td><td>t</td><td>h</td><td>i</td><td>r</td></tr><tr><td>Decrypted letter</td><td>m</td><td>e</td><td>e</td><td>t</td><td>a</td><td>t</td><td>d</td><td>a</td><td>w</td><td>n</td></tr></table>		
	C1	Award 1 mark from: • flmktrwhee mapped to thirtythir (1) • Flmktr mapped to thirty (1)		
	C2	At least one letter decrypted correctly (1)		
	C3	At least one word decrypted correctly (1)		
	C4	Decrypted message 'meet at dawn' (1)		(4)

Type MS · Find named items in a program

Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 3 of 8 · find part d.ii in this mark scheme extract

Question	IMP	Answer	Additional Guidance	Mark
3 (d)(i)	D1	Award 1 mark for: • cipher letter / a single encrypted letter (1)	Do not accept word/message/text	(1)

Type MS · Find named items in a program

Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 4 of 8 · find part d.ii in this mark scheme extract

		• cipherLetter / a single encrypted letter (1)		(1)
3 (d)(ii)	D2	Award 1 mark for any of: • keywordLetter • plaintextLetter	Ignore case	(1)

Type MS · Find named items in a program

Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 5 of 8 · find part d.ii in this mark scheme extract

		• plaintextletter		(1)
3 (d)(iii)	D2	Award 1 mark for any of: • subprogram that is already defined • subprogram that is already written • subprogram that is already compiled • subprogram that can be called without having to write code for it	Must be clear they are referring to subprogram provided by the language itself	(1)

Type MS · Find named items in a program

Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 6 of 8 · find part d.ii in this mark scheme extract

Type MS · Find named items in a program

Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 7 of 8 · find part d.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (a)	A1	Get plaintext and store in plaintext variable	Accept alternative wording Line numbers may be different compared to the examples shown When testing the completed code use lowercase for the input	(1)
	A2	Get key and store in key variable		(1)
	A3	Validate key (must repeat until a valid key has been input)		(1)
	A4	Open file to write/append		(1)
	A5	Write cipher text		(1)
	A6	Close file		(1)
	A7	Displays ciphertext		(1)
	A8	Executing and producing correct output to file and screen (must have A3)		(1)
Code examples				
Python				

```
#       Add your code to get the plaintext and convert it to lowercase
plaintext = input("Enter the plaintext using lowercase letters: ")
plaintext = plaintext.lower()
```

```
# Add your code to get the key and make sure the key is between 1 and 25
while key < 1 or key > 25:
    key = int(input("Enter the key - a number between 1 and 25 "))
```

```
# Add your code to write the ciphertext to a text file
cipherFile = open("Cipher.txt", "w")
cipherFile.write(ciphertext)
cipherFile.close()
```

```
#       Add your code to display the ciphertext
print("The ciphertext message is:",ciphertext)
```

Type MS · Find named items in a program

Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 8 of 8 · find part d.ii in this mark scheme extract

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 1 of 11 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (a) (i)	A1	C# and Python monthNumber/monthsAndDays/count/found (1) Java monthNumber/monthsAndDays/count/found/input		(1)

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 2 of 11 · find part a.i in this mark scheme extract

3 (a) (ii)	A2	monthsAndDays (1)	(1)

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 3 of 11 · find part a.i in this mark scheme extract

3 (a) (iii)	A3	C# • WriteLine, ToString, ToInt32, ReadLine Java • Print, ParseInt, nextLine, ToString, Equals Python • print/len/int/input (1)	(1)

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 4 of 11 · find part a.i in this mark scheme extract

3 (a) (iv)	A4	Display/showMonthNameAndDays (1)	(1)

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 5 of 11 · find part a.i in this mark scheme extract

3 (a) (v)	A5	pMonth (1)	(1)

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 6 of 11 · find part a.i in this mark scheme extract

3 (a) (vi)	A6 A7	**Award 1 mark for each of:** **Type of test data** / **Test data** Boundary (1) / 1 Erroneous (1) / 15		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 7 of 11 · find part a.i in this mark scheme extract

3 (b)	B1 B2	**Award up to 2 marks for a linked explanation such as:** • A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1) • (Using parameters) enables reusable solutions (1) using different data values/arguments (1) • Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1)		(2)

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 8 of 11 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 9 of 11 · find part a.i in this mark scheme extract

3 (c)	**Award 1 mark for each of:**			
	C1	Condition to check whether the year exactly divides by 4 or equivalent condition (1)		
	C2	Condition to check whether year exactly divides by 400 or equivalent condition (1)		
	C3	Condition to check whether year does not exactly divide by 100 or equivalent condition (1)		
	C4	Correct message for at least one of the conditions (1)		
	C5	Program executes and produces the correct output (1)		
	C6	Efficient solution (1)		
				(6)
Code examples				

C#	```
	if (year % 400 == 0)
	{
	Console.WriteLine(year + " is a leap year");
	}
	else if (year % 4 == 0 && year % 100 !=0)
	{
	Console.WriteLine(year + " is a leap year");
	}
	else
	{
	Console.WriteLine(year + " is not a leap year");
	}
	// End of main program
	``			

Java	```
	if (year % 400 == 0)
	{
	System.out.print(year + " is a leap year");
	}
	else if (year % 4 == 0 && year % 100 !=0)
	{
	System.out.print(year + " is a leap year");
	}
	else
	{
	System.out.print(year + " is not a leap year");
	}
	``			

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 10 of 11 · find part a.i in this mark scheme extract

Python	`if year % 400 == 0:`
	`  print(year,"is a leap year")`
	`elif year % 4 == 0 and year % 100 !=0:`
	`  print(year,"is a lear year")`
	`else:`
	`  print(year,"is not a leap year")`

Type MS · Find named items in a program

Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 11 of 11 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 1 of 11 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (a) (i)	A1	C# and Python monthNumber/monthsAndDays/count/found (1) Java monthNumber/monthsAndDays/count/found/input		(1)

Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 2 of 11 · find part a.ii in this mark scheme extract

3 (a) (ii)	A2	monthsAndDays (1)	(1)

Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 3 of 11 · find part a.ii in this mark scheme extract

3 (a) (iii)	A3	C# • WriteLine, ToString, ToInt32, ReadLine Java • Print, ParseInt, nextLine, ToString, Equals Python • print/len/int/input (1)	(1)

Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 4 of 11 · find part a.ii in this mark scheme extract

3 (a) (iv)	A4	Display/showMonthNameAndDays (1)	(1)

Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 5 of 11 · find part a.ii in this mark scheme extract



Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 6 of 11 · find part a.ii in this mark scheme extract

3 (a) (vi)	A6 A7	**Award 1 mark for each of:** **Type of test data** / **Test data** Boundary (1) / 1 Erroneous (1) / 15		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 7 of 11 · find part a.ii in this mark scheme extract

3 (b)	B1 B2	**Award up to 2 marks for a linked explanation such as:** • A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1) • (Using parameters) enables reusable solutions (1) using different data values/arguments (1) • Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1)		(2)

Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 8 of 11 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 9 of 11 · find part a.ii in this mark scheme extract

3 (c)	**Award 1 mark for each of:**			
	C1	Condition to check whether the year exactly divides by 4 or equivalent condition (1)		
	C2	Condition to check whether year exactly divides by 400 or equivalent condition (1)		
	C3	Condition to check whether year does not exactly divide by 100 or equivalent condition (1)		
	C4	Correct message for at least one of the conditions (1)		
	C5	Program executes and produces the correct output (1)		
	C6	Efficient solution (1)		
				(6)
Code examples				
C#				
Java				

```
if (year % 400 == 0)
 {
     Console.WriteLine(year + " is a leap year");
 }
else if (year % 4 == 0 && year % 100 !=0)
 {
     Console.WriteLine(year + " is a leap year");
 }
 else
 {
     Console.WriteLine(year + " is not a leap year");
 }
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 10 of 11 · find part a.ii in this mark scheme extract

Python	`if year % 400 == 0:`
	`  print(year,"is a leap year")`
	`elif year % 4 == 0 and year % 100 !=0:`
	`  print(year,"is a lear year")`
	`else:`
	`  print(year,"is not a leap year")`

Type MS · Find named items in a program

Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 11 of 11 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 1 of 11 · find part a.iii in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
3 (a) (i)	A1	C# and Python monthNumber/monthsAndDays/count/found (1) Java monthNumber/monthsAndDays/count/found/input		(1)

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 2 of 11 · find part a.iii in this mark scheme extract

3 (a) (ii)	A2	monthsAndDays (1)	(1)

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 3 of 11 · find part a.iii in this mark scheme extract

3 (a) (iii)	A3	C# • WriteLine, ToString, ToInt32, ReadLine Java • Print, ParseInt, nextLine, ToString, Equals Python • print/len/int/input (1)	(1)

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 4 of 11 · find part a.iii in this mark scheme extract

3 (a) (iv)	A4	Display/showMonthNameAndDays (1)	(1)

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 5 of 11 · find part a.iii in this mark scheme extract

3 (a) (v)	A5	pMonth (1)	(1)

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 6 of 11 · find part a.iii in this mark scheme extract

3 (a) (vi)	A6 A7	**Award 1 mark for each of:** **Type of test data** / **Test data** Boundary (1) / 1 Erroneous (1) / 15		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 7 of 11 · find part a.iii in this mark scheme extract

3 (b)	B1 B2	**Award up to 2 marks for a linked explanation such as:** • A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1) • (Using parameters) enables reusable solutions (1) using different data values/arguments (1) • Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1)		(2)

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 8 of 11 · find part a.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 9 of 11 · find part a.iii in this mark scheme extract

3 (c)	**Award 1 mark for each of:**			
	C1	Condition to check whether the year exactly divides by 4 or equivalent condition (1)		
	C2	Condition to check whether year exactly divides by 400 or equivalent condition (1)		
	C3	Condition to check whether year does not exactly divide by 100 or equivalent condition (1)		
	C4	Correct message for at least one of the conditions (1)		
	C5	Program executes and produces the correct output (1)		
	C6	Efficient solution (1)		
				(6)
Code examples				
C#				
Java				

```
if (year % 400 == 0)
 {
     Console.WriteLine(year + " is a leap year");
 }
else if (year % 4 == 0 && year % 100 !=0)
 {
     Console.WriteLine(year + " is a leap year");
 }
 else
 {
     Console.WriteLine(year + " is not a leap year");
 }
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 10 of 11 · find part a.iii in this mark scheme extract

Python	`if year % 400 == 0:`
	`  print(year,"is a leap year")`
	`elif year % 4 == 0 and year % 100 !=0:`
	`  print(year,"is a lear year")`
	`else:`
	`  print(year,"is not a leap year")`

Type MS · Find named items in a program

Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 11 of 11 · find part a.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 1 of 11 · find part a.iv in this mark scheme extract

Question		Answer	Additional Guidance	Mark
3 (a) (i)	A1	C# and Python monthNumber/monthsAndDays/count/found (1) Java monthNumber/monthsAndDays/count/found/input		(1)

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 2 of 11 · find part a.iv in this mark scheme extract

3 (a) (ii)	A2	monthsAndDays (1)	(1)

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 3 of 11 · find part a.iv in this mark scheme extract

3 (a) (iii)	A3	C# • WriteLine, ToString, ToInt32, ReadLine Java • Print, ParseInt, nextLine, ToString, Equals Python • print/len/int/input (1)	(1)

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 4 of 11 · find part a.iv in this mark scheme extract

3 (a) (iv)	A4	Display/showMonthNameAndDays (1)	(1)

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 5 of 11 · find part a.iv in this mark scheme extract

3 (a) (v)	A5	pMonth (1)	(1)

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 6 of 11 · find part a.iv in this mark scheme extract

3 (a) (vi)	A6 A7	**Award 1 mark for each of:** **Type of test data** / **Test data** Boundary (1) / 1 Erroneous (1) / 15		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 7 of 11 · find part a.iv in this mark scheme extract

3 (b)	B1 B2	**Award up to 2 marks for a linked explanation such as:** • A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1) • (Using parameters) enables reusable solutions (1) using different data values/arguments (1) • Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1)		(2)

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 8 of 11 · find part a.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 9 of 11 · find part a.iv in this mark scheme extract

3 (c)	**Award 1 mark for each of:**			
	C1	Condition to check whether the year exactly divides by 4 or equivalent condition (1)		
	C2	Condition to check whether year exactly divides by 400 or equivalent condition (1)		
	C3	Condition to check whether year does not exactly divide by 100 or equivalent condition (1)		
	C4	Correct message for at least one of the conditions (1)		
	C5	Program executes and produces the correct output (1)		
	C6	Efficient solution (1)		
				(6)
Code examples				
C#				
Java				

```
if (year % 400 == 0)
{
    Console.WriteLine(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    Console.WriteLine(year + " is a leap year");
}
else
{
    Console.WriteLine(year + " is not a leap year");
}
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 10 of 11 · find part a.iv in this mark scheme extract

Python	`if year % 400 == 0:`
	`  print(year,"is a leap year")`
	`elif year % 4 == 0 and year % 100 !=0:`
	`  print(year,"is a lear year")`
	`else:`
	`  print(year,"is not a leap year")`

Type MS · Find named items in a program

Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 11 of 11 · find part a.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 1 of 11 · find part a.v in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
3 (a) (i)	A1	C# and Python monthNumber/monthsAndDays/count/found (1) Java monthNumber/monthsAndDays/count/found/input		(1)

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 2 of 11 · find part a.v in this mark scheme extract

3 (a) (ii)	A2	monthsAndDays (1)	(1)

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 3 of 11 · find part a.v in this mark scheme extract

3 (a) (iii)	A3	C# • WriteLine, ToString, ToInt32, ReadLine Java • Print, ParseInt, nextLine, ToString, Equals Python • print/len/int/input (1)	(1)

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 4 of 11 · find part a.v in this mark scheme extract

3 (a) (iv)	A4	Display/showMonthNameAndDays (1)	(1)

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 5 of 11 · find part a.v in this mark scheme extract

3 (a) (v)	A5	pMonth (1)	(1)

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 6 of 11 · find part a.v in this mark scheme extract

3 (a) (vi)	A6 A7	**Award 1 mark for each of:** **Type of test data** / **Test data** Boundary (1) / 1 Erroneous (1) / 15		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 7 of 11 · find part a.v in this mark scheme extract

3 (b)	B1 B2	**Award up to 2 marks for a linked explanation such as:** • A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1) • (Using parameters) enables reusable solutions (1) using different data values/arguments (1) • Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1)		(2)

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 8 of 11 · find part a.v in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 9 of 11 · find part a.v in this mark scheme extract

3 (c)	**Award 1 mark for each of:**			
	C1	Condition to check whether the year exactly divides by 4 or equivalent condition (1)		
	C2	Condition to check whether year exactly divides by 400 or equivalent condition (1)		
	C3	Condition to check whether year does not exactly divide by 100 or equivalent condition (1)		
	C4	Correct message for at least one of the conditions (1)		
	C5	Program executes and produces the correct output (1)		
	C6	Efficient solution (1)		
				(6)
Code examples				

C#	```
	if (year % 400 == 0)
	{
	Console.WriteLine(year + " is a leap year");
	}
	else if (year % 4 == 0 && year % 100 !=0)
	{
	Console.WriteLine(year + " is a leap year");
	}
	else
	{
	Console.WriteLine(year + " is not a leap year");
	}
	// End of main program
	``			

Java	```
	if (year % 400 == 0)
	{
	System.out.print(year + " is a leap year");
	}
	else if (year % 4 == 0 && year % 100 !=0)
	{
	System.out.print(year + " is a leap year");
	}
	else
	{
	System.out.print(year + " is not a leap year");
	}
	``			

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 10 of 11 · find part a.v in this mark scheme extract

Python	`if year % 400 == 0:`
	`  print(year,"is a leap year")`
	`elif year % 4 == 0 and year % 100 !=0:`
	`  print(year,"is a lear year")`
	`else:`
	`  print(year,"is not a leap year")`

Type MS · Find named items in a program

Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 11 of 11 · find part a.v in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 1 of 7 · find part a.iv in this mark scheme extract



Type MS · Find named items in a program

Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 2 of 7 · find part a.iv in this mark scheme extract



Type MS · Find named items in a program

Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 3 of 7 · find part a.iv in this mark scheme extract



Type MS · Find named items in a program

Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 4 of 7 · find part a.iv in this mark scheme extract



Type MS · Find named items in a program

Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 5 of 7 · find part a.iv in this mark scheme extract

2 (a)(v)	A5	pNumber		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 6 of 7 · find part a.iv in this mark scheme extract

2 (b)	**Award 1 mark for each of:**		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the input message.	(11)
	Evidence should be found in the Function			
	B1	Function interface is correct (1)		
	B2	Condition correct (for the number 1) (1)		
	B3	Loop initialised correctly (from 2 to the input number) (1)		
	B4	If condition correct (modulus) (1)		
	B5	Check returned (1)		
	Evidence should be found in the Main program			
	B6	Input of number stored as an integer variable (1)		
	B7	Input includes a space/colon / new line before the input value		
	B8	Function called correctly (1)		
	B9	Display includes number and message for either a prime number **OR** not (1)		
	Overall			
	B10	Compiling without syntax errors (1)		
	B11	Executing and producing the correct output (1)		
Code examples				
Python				

```python
# Write the function here
def checkPrime(pNumber):
    if pNumber == 1:
        check = False
    else:
        check = True
        for count in range(2,pNumber):
            if pNumber % count == 0 :
                check = False
    return check

# Write the main program here
number = int(input("Enter a number: "))
result = checkPrime(number)
if result == True:
    print(number,"is a prime number")
else:
    print(number,"is not a prime number")
```

Type MS · Find named items in a program

Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 7 of 7 · find part a.iv in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Find named items in a program

Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 1 of 7 · find part a.v in this mark scheme extract



Type MS · Find named items in a program

Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 2 of 7 · find part a.v in this mark scheme extract



Type MS · Find named items in a program

Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 3 of 7 · find part a.v in this mark scheme extract



Type MS · Find named items in a program

Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 4 of 7 · find part a.v in this mark scheme extract



Type MS · Find named items in a program

Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 5 of 7 · find part a.v in this mark scheme extract

2 (a)(v)	A5	pNumber		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 6 of 7 · find part a.v in this mark scheme extract

2 (b)	**Award 1 mark for each of:**		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the input message.	(11)
	Evidence should be found in the Function			
	B1	Function interface is correct (1)		
	B2	Condition correct (for the number 1) (1)		
	B3	Loop initialised correctly (from 2 to the input number) (1)		
	B4	If condition correct (modulus) (1)		
	B5	Check returned (1)		
	Evidence should be found in the Main program			
	B6	Input of number stored as an integer variable (1)		
	B7	Input includes a space/colon / new line before the input value		
	B8	Function called correctly (1)		
	B9	Display includes number and message for either a prime number **OR** not (1)		
	Overall			
	B10	Compiling without syntax errors (1)		
	B11	Executing and producing the correct output (1)		
Code examples				
Python				

```python
# Write the function here
def checkPrime(pNumber):
    if pNumber == 1:
        check = False
    else:
        check = True
        for count in range(2,pNumber):
            if pNumber % count == 0 :
                check = False
    return check

# Write the main program here
number = int(input("Enter a number: "))
result = checkPrime(number)
if result == True:
    print(number,"is a prime number")
else:
    print(number,"is not a prime number")
```

Type MS · Find named items in a program

Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 7 of 7 · find part a.v in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Find named items in a program

Answer 14 of 18 · Question 1 part b.iii · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part b.iii in this mark scheme extract

1(a)	A1 A2 A3	Award **one** mark for each correct cell: Input: Debit / Credit / Description Processing: Balance + Credit / Balance - Debit Output: Balance	Input: accept examples of inputs e.g. 10.23 (must be in Figure 1) Processing; mark first 3 things (ignore extra text) **if** correct e.g. **balance + credit** – debit (1) **balance + credit** + debit (1) balance – credit (0) accept examples from table e.g. 128.35 – 10.23 (1) Output: accept description of formula e.g. subtract debit from balance (1)	**3**

Question	mp	Answer	Additional Guidance	Mark
		Award **one** mark for each of:		
1(b)(i)	B1	This is 5%		**1**
1(b)(ii)	B2	or Accept || for C# and Java		**1**
1(b)(iii)	B3	grossAmt netAmt		**1**
1(b)(iv)	B4	TAX		**1**
1(b)(v)	B5	inGross		**1**

Type MS · Find named items in a program

Answer 14 of 18 · Question 1 part b.iii · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part b.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1(c)(i)	C1	Award **one** mark for: To store a value that will not change (during program execution) (1)		**1**
1(c)(ii)	C2	Award **one** mark for any of the following: • It is less likely to be changed by accident (translation will error if attempt assignment)(1) • The value only needs to be changed in one place (for maintenance) (1) • It is easier to read/understand **the logic** of (1) • Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1) • Can make code more efficient (optimisation / value stored rather than reference)	MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user.	**1**

Type MS · Find named items in a program

Answer 14 of 18 · Question 1 part b.iii · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part b.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 15 of 18 · Question 1 part b.iv · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part b.iv in this mark scheme extract

1(a)	A1 A2 A3	Award **one** mark for each correct cell: Input: Debit / Credit / Description Processing: Balance + Credit / Balance - Debit Output: Balance	Input: accept examples of inputs e.g. 10.23 (must be in Figure 1) Processing; mark first 3 things (ignore extra text) **if** correct e.g. **balance + credit** – debit (1) **balance + credit** + debit (1) balance – credit (0) accept examples from table e.g. 128.35 – 10.23 (1) Output: accept description of formula e.g. subtract debit from balance (1)	**3**

Question	mp	Answer	Additional Guidance	Mark
		Award **one** mark for each of:		
1(b)(i)	B1	This is 5%		**1**
1(b)(ii)	B2	or Accept || for C# and Java		**1**
1(b)(iii)	B3	grossAmt netAmt		**1**
1(b)(iv)	B4	TAX		**1**
1(b)(v)	B5	inGross		**1**

Type MS · Find named items in a program

Answer 15 of 18 · Question 1 part b.iv · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part b.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1(c)(i)	C1	Award **one** mark for: To store a value that will not change (during program execution) (1)		**1**
1(c)(ii)	C2	Award **one** mark for any of the following: • It is less likely to be changed by accident (translation will error if attempt assignment)(1) • The value only needs to be changed in one place (for maintenance) (1) • It is easier to read/understand **the logic** of (1) • Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1) • Can make code more efficient (optimisation / value stored rather than reference)	MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user.	**1**

Type MS · Find named items in a program

Answer 15 of 18 · Question 1 part b.iv · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part b.iv in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 16 of 18 · Question 1 part b.v · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part b.v in this mark scheme extract

1(a)	A1 A2 A3	Award **one** mark for each correct cell: Input: Debit / Credit / Description Processing: Balance + Credit / Balance - Debit Output: Balance	Input: accept examples of inputs e.g. 10.23 (must be in Figure 1) Processing; mark first 3 things (ignore extra text) **if** correct e.g. **balance + credit** – debit (1) **balance + credit** + debit (1) balance – credit (0) accept examples from table e.g. 128.35 – 10.23 (1) Output: accept description of formula e.g. subtract debit from balance (1)	**3**

Question	mp	Answer	Additional Guidance	Mark
		Award **one** mark for each of:		
1(b)(i)	B1	This is 5%		**1**
1(b)(ii)	B2	or Accept || for C# and Java		**1**
1(b)(iii)	B3	grossAmt netAmt		**1**
1(b)(iv)	B4	TAX		**1**
1(b)(v)	B5	inGross		**1**

Type MS · Find named items in a program

Answer 16 of 18 · Question 1 part b.v · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part b.v in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
1(c)(i)	C1	Award **one** mark for: To store a value that will not change (during program execution) (1)		**1**
1(c)(ii)	C2	Award **one** mark for any of the following: • It is less likely to be changed by accident (translation will error if attempt assignment)(1) • The value only needs to be changed in one place (for maintenance) (1) • It is easier to read/understand **the logic** of (1) • Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1) • Can make code more efficient (optimisation / value stored rather than reference)	MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user.	**1**

Type MS · Find named items in a program

Answer 16 of 18 · Question 1 part b.v · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part b.v in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 17 of 18 · Question 1 part d.i · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part d.i in this mark scheme extract

1(a)	A1	**The only correct answer is B** ***A*** *is not correct because for is used in iteration* ***C*** *is not correct because new is used to declare an object* ***D*** *is not correct because while is used in repetition*		**1**

Question	mp	Answer	Additional Guidance	Mark
1(b)(i)	B1 B2 B3	Award one mark for each correct cell:		**3**

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.		✔	
The program does not translate.	✔		
The program crashes during execution.			✔

1 of 23

Type MS · Find named items in a program

Answer 17 of 18 · Question 1 part d.i · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part d.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(b)(ii)	B4	Line 15 - : added to end of line		
	B5	Line 20 – 0 changed to 1		
	B6	Line 22 – missing comma added between string and greenCount		3

```
1  # Q01bii
2
3  # initialise counters
4  ballCount = 0
5  greenCount = 0
6  redCount = 0
7
8  ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
9              "green", "green", "green", "red", "green", "red", "green", "red",
10             "red", "green", "green", "red", "red", "green", "green", "red",
11             "green", "red", "red", "green", "green", "red", "green", "green",
12             "red", "green", "green", "green", "red", "green", "red","green"]
13
14 # iterate over ball array
15 for ball in ballArray:
16     ballCount = ballCount + 1
17     if ball == "green":
18         greenCount = greenCount +1
19     else:
20         redCount = redCount + 1
21
22 print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
23       "balls \n", "Total: ", ballCount, "balls")
```

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Type MS · Find named items in a program

Answer 17 of 18 · Question 1 part d.i · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part d.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(c)(i)	C1	Award one mark for: • To store a value that must not be changed/prevent accidental change of value • To simplify code maintenance/so that if the value must be updated the change only has to made in a single place • To avoid inconsistency error of specifying same value multiple times • To make code more readable/give meaningful names to values	Accidental change cannot refer to 'user'	1

Question	mp	Answer	Additional guidance	Mark
1(c)(ii)	C2 C3	Award **two** marks for a linked description such as: • A variable stores a single value (1) whereas an array stores multiple values (1)		2

Question	mp	Answer	Additional guidance	Mark
1(d)(i)	D1	Award **one** mark for: • getWages (1)	Accept 'main' Accept: def getWages def getWages (inSales, inAge) line 7 (Python)	1

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Type MS · Find named items in a program

Answer 17 of 18 · Question 1 part d.i · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part d.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(d)(ii)	D2	Award **one** mark for: • and (1)	Do not accept line numbers Accept: &&	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iii)	D3	Award **one** mark for any of the following: • sales (1) • wages (1) • inSales (1)	Do not accept line numbers	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iv)	D4	Award **one** mark for any of the following: • inSales (1) • inAge (1)	Do not accept line numbers	1

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Type MS · Find named items in a program

Answer 17 of 18 · Question 1 part d.i · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part d.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark

Type MS · Find named items in a program

Answer 18 of 18 · Question 1 part d.iii · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part d.iii in this mark scheme extract

1(a)	A1	**The only correct answer is B** ***A*** *is not correct because for is used in iteration* ***C*** *is not correct because new is used to declare an object* ***D*** *is not correct because while is used in repetition*		**1**

Question	mp	Answer	Additional Guidance	Mark
1(b)(i)	B1 B2 B3	Award one mark for each correct cell:		**3**

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.		✔	
The program does not translate.	✔		
The program crashes during execution.			✔

1 of 23

Type MS · Find named items in a program

Answer 18 of 18 · Question 1 part d.iii · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part d.iii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(b)(ii)	B4	Line 15 - : added to end of line		
	B5	Line 20 – 0 changed to 1		
	B6	Line 22 – missing comma added between string and greenCount		3

```
# Q01bii

# initialise counters
ballCount = 0
greenCount = 0
redCount = 0

ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
             "green", "green", "green", "red", "green", "red", "green", "red",
             "red", "green", "green", "red", "red", "green", "green", "red",
             "green", "red", "red", "green", "green", "red", "green", "green",
             "red", "green", "green", "green", "red", "green", "red","green"]

# iterate over ball array
for ball in ballArray:
    ballCount = ballCount + 1
    if ball == "green":
        greenCount = greenCount +1
    else:
        redCount = redCount + 1

print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
      "balls \n", "Total: ", ballCount, "balls")
```

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Type MS · Find named items in a program

Answer 18 of 18 · Question 1 part d.iii · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part d.iii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(c)(i)	C1	Award one mark for: • To store a value that must not be changed/prevent accidental change of value • To simplify code maintenance/so that if the value must be updated the change only has to made in a single place • To avoid inconsistency error of specifying same value multiple times • To make code more readable/give meaningful names to values	Accidental change cannot refer to 'user'	1

Question	mp	Answer	Additional guidance	Mark
1(c)(ii)	C2 C3	Award **two** marks for a linked description such as: • A variable stores a single value (1) whereas an array stores multiple values (1)		2

Question	mp	Answer	Additional guidance	Mark
1(d)(i)	D1	Award **one** mark for: • getWages (1)	Accept 'main' Accept: def getWages def getWages (inSales, inAge) line 7 (Python)	1

3 of 23

Type MS · Find named items in a program

Answer 18 of 18 · Question 1 part d.iii · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part d.iii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(d)(ii)	D2	Award **one** mark for: • and (1)	Do not accept line numbers Accept: &&	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iii)	D3	Award **one** mark for any of the following: • sales (1) • wages (1) • inSales (1)	Do not accept line numbers	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iv)	D4	Award **one** mark for any of the following: • inSales (1) • inAge (1)	Do not accept line numbers	1

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Type MS · Find named items in a program

Answer 18 of 18 · Question 1 part d.iii · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part d.iii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark